

# Beyond the Colony-Forming-Unit: Rapid Bacterial Evaluation in Osteomyelitis

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## Abstract

Examination of bacteria/host cell interactions is important for understanding the aetiology of many infectious diseases. The colony-forming-unit (CFU) has been the standard for quantifying bacterial burden for the past century, however, this suffers from low sensitivity and is dependent on bacterial culturability *in vitro*. Our data demonstrate the discrepancy between the CFU and bacterial genome copy number in an osteomyelitis-relevant co-culture system and we confirm diagnosis and quantify bacterial load in clinical bone specimens. This study provides an improved workflow for the quantification of bacterial burden in such cases.

### eLife assessment

This **fundamental** study addresses discrepancies in determining bacterial burden in osteomyelitis as determined by culture and enumeration using DNA. The authors present **compelling** data demonstrating the emergence of discrepancies between CFU counts and genome copy numbers detected by PCR in *Staphylococcus aureus* strains infecting osteocyte-like cells. The observations represent a substantial addition to the field of musculoskeletal infection, with possible broad applicability and clinical benefit to other infectious diseases.

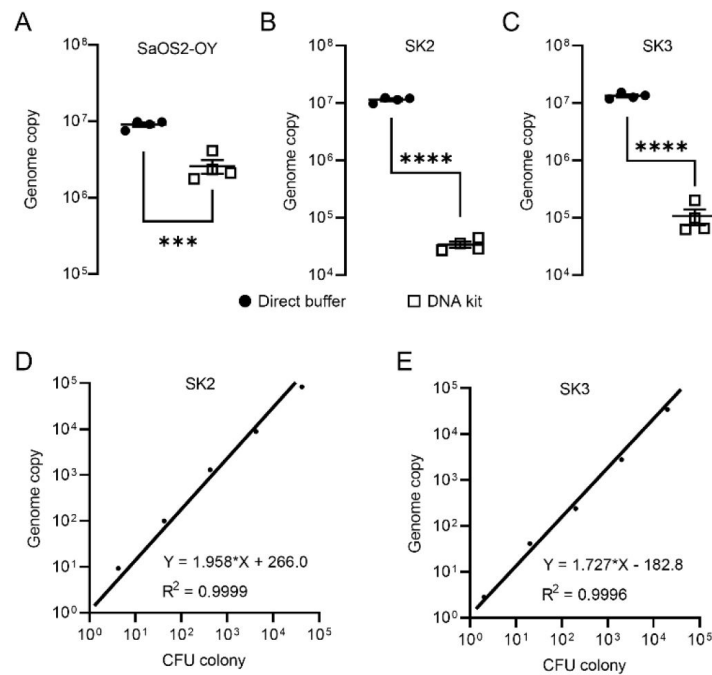
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Co-culturing mammalian host cells with bacteria *in vitro* is an important tool for examining the host-pathogen interaction in infectious disease modelling [1 ]. One of the essential readouts is the intracellular bacterial burden carried by the infected host cells. The most commonly used method for enumerating such bacterial load is counting the colony-forming-unit (CFU) number by agar plate culturing following the lysis of host cells and release of bacteria [2 ]. However, the reliability of such measurements is impacted by various factors, including, but not limited to, host cell/tissue type, bacterial strain, bacterial load, and culturing conditions for CFU enumeration [3 , 4 ]. Osteomyelitis (OM), an infectious disease with pathogen-mediated infection in bone

tissues, features a considerable proportion of infected but culture-negative cases, ranging between 20-40% [5, 7]. This clear deficiency in CFU evaluation create difficulties for the accurate diagnosis and therefore treatment of OM.

*Staphylococci*, including coagulase-negative species, such as *S. epidermidis*, and *S. aureus*, comprise the major pathogens in adult OM [8]. In this study, we employed a previously established osteocyte-like cell model, differentiated SaOS2 (SaOS2-OY) and the single most common causative pathogen in OM, *S. aureus*, to simulate the *in vitro* infection of osteocytes, the most abundant cell type in bone tissue [9, 10]. Two previously characterised *S. aureus* strains, a high virulence (methicillin-resistant *S. aureus*; MRSA) strain, WCH-SK2 (SK2), shown previously to establish an intracellular infection in human osteocytes [11] and a low virulence (methicillin-sensitive *S. aureus*; MSSA) strain, WCH-SK3 (SK3) [12], were chosen for experimentation. In addition to CFU enumeration to quantify bacterial number, a polymerase chain reaction (PCR) approach measuring genomic DNA copies represented by a single copy gene within the *S. aureus* genome, was also performed in parallel. In the past, real time quantitative PCR (qPCR) assays have been widely used for the quantification of bacteria. However, the working principle of standard qPCR is dependent on the establishment of a standard curve using a known number of bacterial gene copies and the fitting of test samples into a linear regression to estimate the bacterial number for each assay [13, 14]. Here, we utilised digital droplet PCR (ddPCR) for this measurement. By its nature, the latter technique offers the advantage of counting the absolute target sequence copy number within the sample and removes the need for normalisation [15]. For all PCR-based assays, the quality of DNA containing the target sequence for amplification and quantification is critical for the sensitivity and reproducibility of readout. Therefore, methods for DNA preparation from both human and bacterial cells were examined. Here, we introduced the usage of DirectPCR™ Lysis Reagent (Direct buffer), a lysis buffer that maximises the release of genomic DNA from samples and is compatible with the downstream PCR analysis from raw cell culture lysate without any purification step. With the application of Direct buffer for DNA preparations in comparison to a commonly used DNA extraction spin column kit (DNA kit), the absolute bacterial genome copy number quantified by ddPCR was measured to be 5-fold higher in SaOS2-OY samples and 100-fold higher in samples from two strains of *S. aureus*, SK2 and SK3; better overall reproducibility among biological replicates was also achieved using Direct buffer over the standard approach (Fig. 1A-C). Serial 10-fold dilutions of bacterial log-phase suspension cultures were used to generate DNA lysates for testing the extent of DNA release. Plotting genome copy number against CFU from the same cultures showed near perfect linear regression relationships (Fig. 1D-E), confirming the complete and highly reproducible release of bacterial DNA. With this validating readout, we sought to determine if the elimination of purification steps by using the Direct buffer might minimise the sample loss associated with the utilisation of conventional column-based extraction methods [16]. Further, the release of genomic material by this reagent appeared to be complete and unbiased in our experiments, which led to higher consistency of inter- and intra-assays.

Using the combined approach of direct lysis DNA preparation and ddPCR quantification, we then examined the *S. aureus* persistence in the intracellular environment in infected SaOS2-OY cells. Following infection, the SaOS2-OY and *S. aureus* co-cultures were lysed to break down host cell structures for the release of intracellular *S. aureus* for CFU development on agar plates. Consistent with previous findings, significant reductions ranging between 10<sup>3</sup>-fold and 10<sup>6</sup>-fold in cultured CFU on agar plates over the 5-day co-culture time period were observed in both SK2 and SK3 infected SaOS2-OY groups (Fig. 2 A-F). This trend was also found in low multiplicities of infection (MOI) (~10<sup>7</sup> total CFU/well, Fig. 2 C & E) and high MOI (~10<sup>8</sup> total CFU/well, Fig. 2 D & F) for initial bacteria-inoculated groups with either strain. However, when total DNA preparations from the whole culture lysates were taken for ddPCR quantification of bacterial genome copy number, differences in the dynamic changes of bacterial load were found. In the



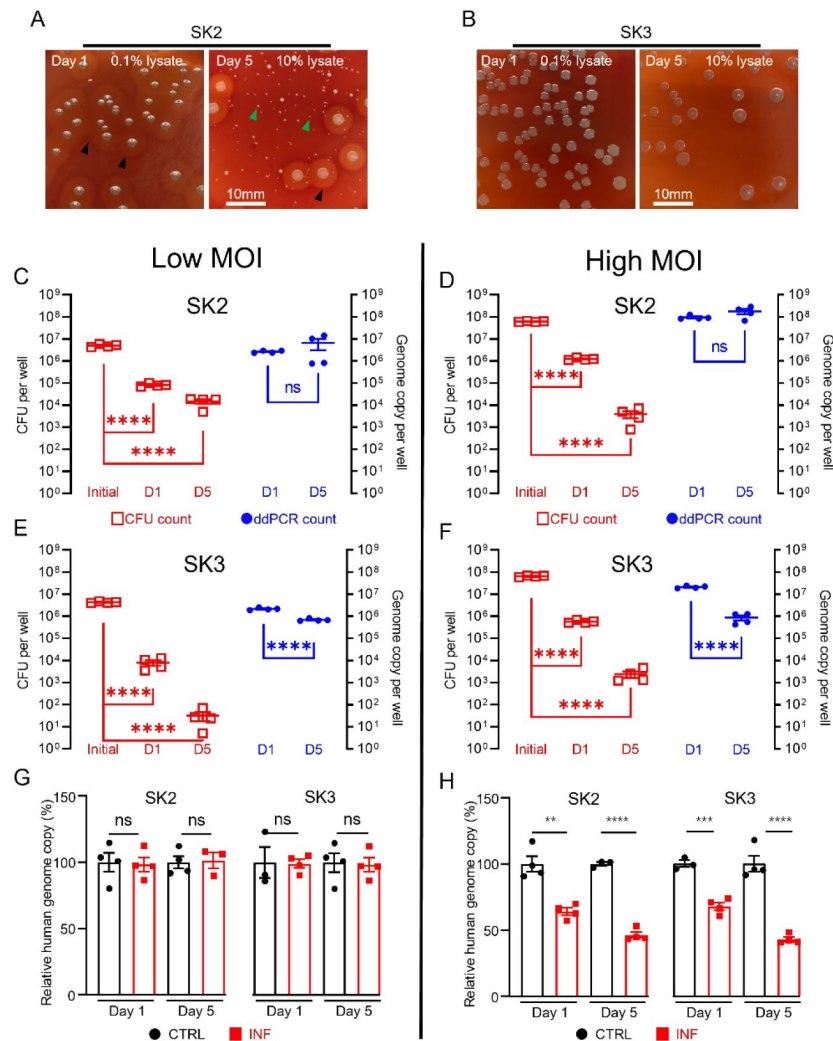
**Fig. 1**

ddPCR genome counting of SaOS2-OY (A), SK2 (B) and SK3 (C), comparing the Direct buffer approach (•) and a standard DNA kit (□) (4 biological replicates with mean and standard errors were shown on graphs, \*\*\* $p < 0.001$  and \*\*\*\* $p < 0.0001$ ); demonstration of complete bacterial chromatin release from SK2 (D) and SK3 (E): correlation with CFU plating. Each data point represented the comparison of one CFU recovery and one total DNA measurement; three independent experiments were carried out for each strain and similar results were achieved; representative results of one experiment are presented.

high virulence strain SK2-infected group, the genome copy numbers by 5 days post-infection were maintained at the levels of  $10^7$  and  $10^8$  cells/well for the low and high MOI groups, respectively (Fig. 2 C & D), similar to the original input levels.

However, the low virulence strain SK3-infected group demonstrated a decrease in genome copy counts. Interestingly, regardless of initial input amount, by 5 days post-infection, the remaining SK3 genome counts dropped to approximately  $10^6$  cells/well for both groups (Fig. 2 E & F). The quantification of human genome copy number was also performed in all groups to estimate the remaining host cells in culture and this also served as an indicator of host cell viability. With the low MOI co-cultures, the human genome copy counts in the presence of either strain of *S. aureus* were not statistically different from uninfected controls, across 1- and 5-days post-infection. When the initial bacterial inoculum was increased 10-fold to  $10^8$  cells per well, reductions in the human genome copies of ~30% and 50% were measured in 1- and 5-day post-infection groups, respectively; such a trend was found to be similar in both SK2- and SK3-infected groups. Overall, our observations demonstrated profound differences in the readout of the presence of intracellular bacteria when using CFU and ddPCR quantification methods. Such inconsistency between these readouts in our model was as high as  $10^6$ -fold, with interaction of factors including initial bacterial load and strain variation. In contrast, comparison of the same quantification methods performed on bacteria suspension cultures yielded differences under 2-fold, indicated by the respective slope factor (less than 2) in the linear regression curves for both SK2 and SK3 strains (see Fig. 1 D & E). Together, our results indicated the dramatic change in bacterial culturability when comparing growth in ideal microbial suspension culture conditions and the growth limiting intracellular environment. Therefore, we propose that the interaction between the host cell response to infection and pathogen adaptation to the intracellular environment will compromise the reliability of the standard CFU counting method alone for the evaluation of bacterial persistence in a bacteria/host cell co-culture experiment. However, phenotypic variations were also observed in the recovered colonies from agar plating, with variations in haemolytic activity and small colony variant (SCV) formation, particularly in the SK2 group (Fig. 2A). SCV variants of *S. aureus* feature reduced metabolic activity and growth but are an indication of adaptation and the establishment of a chronic infection. This is consistent with the findings of reductions in CFU recovery but maintained levels of detectable genome copies, at least in the SK2-infected group.

To examine the applicability of this workflow to a clinical setting, we next extended the tests to clinical human bone specimens. Three clinically culture-negative periprosthetic joint infection (PJI) cases were chosen for analysis. All three cases were confirmed infected by clinical observation for typical PJI symptoms by clinicians, using the current gold-standard Musculoskeletal Infection Society (MSIS) criteria [17]; while each patient presented with a sinus tract communicating with their prosthesis, a major diagnostic criterion, neither the culturing of the synovial membrane biopsies nor synovial fluid around the prosthesis returned any positive cultures from a clinical laboratory. The bone tissue sections from the 3 patients were examined by Masson's trichrome staining in comparison to tissue section from an osteoarthritis (OA) patient as non-infection control, to examine the bone matrix degradation, according to our previous study [18]. With this method, the red colour indicates intact bone matrix collagen and the blue colour corresponds to degraded collagen. Within all the histological observations, matrix degradation was found in sections of all three PJI patients (Fig. 3 B-D), whereas the staining of the control OA patient bone indicated intact collagen (Fig. 3 A), consistent with infection in the 3 PJI patient specimens [18]. In addition to the gold-standard diagnostic approach and histological staining, we performed CFU analysis using homogenised bone tissue, and negative culture results were also recorded. In this study, the samples containing total genomic DNA were prepared from histological sections in conjunction with the use of the DirectPCR™ Lysis reagent. The prepared DNA samples were put through a PCR amplification using primer set targeting a highly conserved bacterial genomic region, elongation factor Tu (*tuf*) [19], for the enrichment of bacterial signal within host/pathogen genomic mixture. The generated amplicons were then sequenced using an Oxford



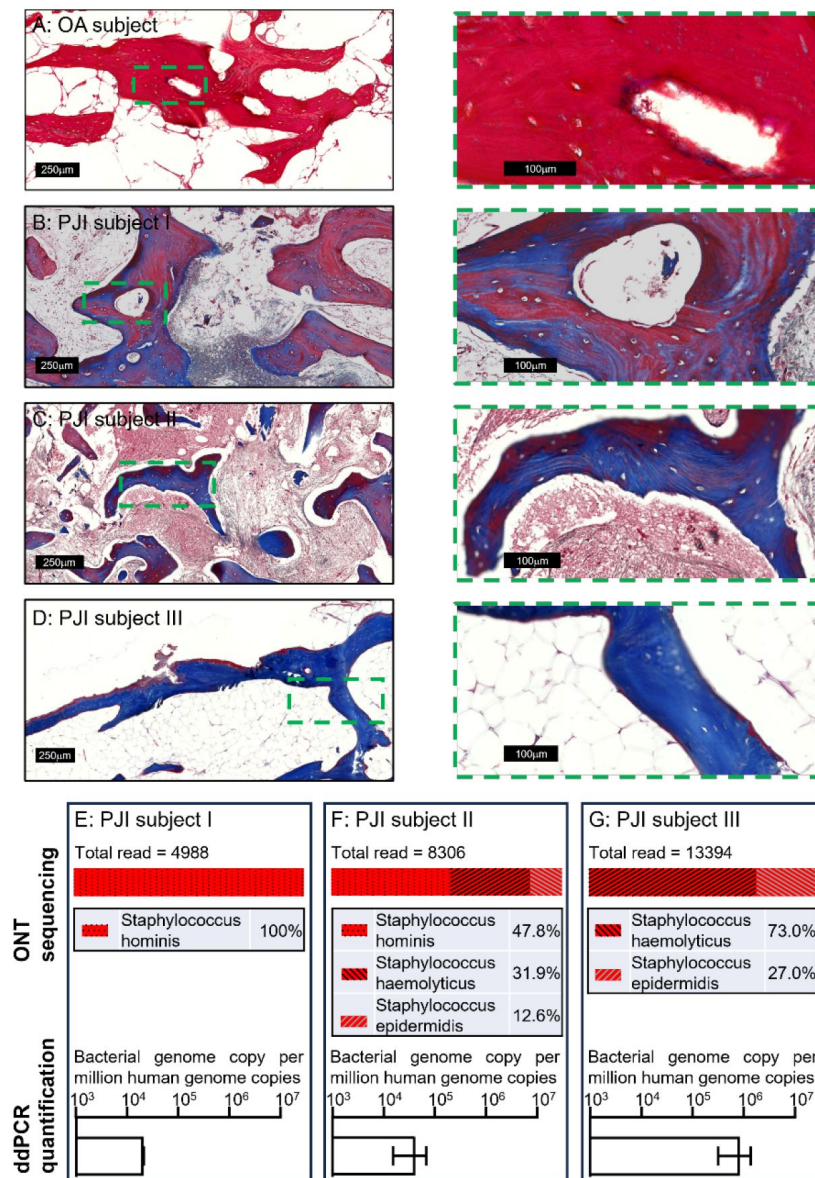
**Fig. 2**

CFU recovery of SK2 (A) and SK3 (B) from host SaOS2-OY cells, with haemolysis reactions (▲) and numerous small colony variants (▲) shown in SK2 group; quantification of SK2 and SK3 using CFU count (□, left Y) and ddPCR count (●, right Y) from low (C & E) and high (D & F) MOI groups; relative human genome copy (%) with CTRL (□) VS INF (■) quantified by ddPCR from low (G) and high (H) MOI groups. (4 biological replicates with mean and standard errors are shown; \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ , \*\*\*\*  $p < 0.0001$ ; ns = not significant)

Nanopore Technology (ONT) Minion sequencer for the identification of bacterial species. All three patient bone DNA samples were confirmed to be bacterial genomic positive, with combinations identified of the coagulase-negative staphylococcal species *S. haemolyticus*, *S. hominis* and *S. epidermidis* (Fig. 3 E-G). The respective bacterial load for each of the samples was then quantified by ddPCR. Each bone sample examined contained between  $2 \times 10^4$  and  $1 \times 10^6$  bacterial genomic copies per million human genomic copies (Fig. 3 E-G). To confirm that the detected bacteria species were not introduced during the handling procedures, bone specimens from five primary total hip replacement (non-infected) patients were processed at the same time along with the PJI specimens; PCR reactions with negative *tuf* amplification for these bone samples indicated that operational contamination was unlikely (Supplementary Fig. 1).

In this study, by the analysis of genomic material, we have addressed the ineffectiveness of the classical bacterial CFU development approach in determining and quantifying a confirmed infection by both an *in vitro* experimental model and by examining human clinical cases. With the introduction of a direct DNA release approach, less handling is required while delivering increased accuracy. The addition of ddPCR provided the advantage of achieving absolute quantification without the need for a PCR standard curve, which is both cost and time effective. For the purposes of unknown pathogen diagnosis in clinical cases, the exact bacterial species readout is required from sequencing the generated amplicons. Here, the utilisation of Oxford Nanopore technology dramatically lowered the equipment and skill demands for performing the analysis, since the MinION sequencer is a portable device and the sequencing results are analysed by the on-board software with little bioinformatics requirement. Noted here, the usual whole genome sequencing approach might not be suitable for this application, as in many of the clinical specimens, the amount of human genome can be orders of magnitude higher than the level of bacterial genomic material. Sequencing and analysing relatively rare bacterial DNA in this scenario is considerably more lengthy and costly. Therefore, a PCR approach was chosen to amplify the bacterial signal for detection. In this study, bacterial gene *tuf* was chosen as the target sequence to generate PCR amplicons instead of the commonly used 16S rRNA gene sequence, because the prior one has been shown greater discriminatory power than 16S [20]. Here, we took *tuf* sequences from *S. aureus* and the three coagulase-negative staphylococcal species identified in the clinical specimens for cross-comparisons (Supplementary Fig. 2A). Within the designated amplified sequence of only 271 bp, 11-22 (or 4.1%-8.1% differences) mismatches were identified between these four species (Supplementary Fig. 2B). For achieving such equivalent differences in 16S sequences among the four strains, PCR amplicons with over 1000 bp in length are required (data not shown). Again, the strategy of analysing a short amplicon not only offers reduced reaction time, more importantly, it minimises the chance of generating non-specific products by shortening the elongation step. The listed primer set targeting the *tuf* gene was confirmed to cover most of the species in genus *Staphylococcus in silico*. In practice, it is challenging to find universal primers that cover the entire bacterial domain. Hence, for future real-world tests in clinical diagnosis, the strategy of using multiple primer sets to cover common PJI pathogens might need to be considered. Together, this workflow is potentially applicable as a point-of-care diagnostic method with the prospect of a rapid turnover: within hours comparing to the current procedure in the magnitude of days.

To conclude, for the purpose of quantifying the intracellular bacterial load in a host-pathogen co-culture setting, we have developed a workflow with the advantages of: a) rapid and labour-saving, with the elimination all the DNA extraction steps by virtue of using the direct lysis approach; b) minimisation of potential sample loss encountered with conventional DNA isolation methods, enabling better accuracy and reproducibility; c) absolute number quantification by ddPCR for achieving superior consistency of intra-assay experiments; d) sequencing analysis using the ONT MinION platform has the potential to become a point-of-care diagnostic approach when performing unknown pathogen identification in a clinical setting. Nevertheless, we suggest that the CFU plating method should not be omitted as it allows the evaluation of bacterial phenotypic adaptation in such experimentation.



**Fig. 3**

Masson's trichrome staining on bone tissue sections of an osteoarthritis (OA) subject (A) and culture negative PJI subjects I-III (B-D). Pathogen profiling using total DNA of bone specimens from the above 3 PJI patients (E-G) by ONT sequencing for the readout of pathogen species and ddPCR to quantify bacterial load as a ratio of bacterial : human genomic copies (error bars shown are the combination of the standard error of the mean from two individual DNA preparations and the device generated error from each of the ddPCR runs, with over 15,000 droplets read per sample).

## Materials and methods

### SaOS2 cell culture and differentiation

The human osteosarcoma cell line SaOS2 differentiated to an osteocyte-like stage was employed for performing the bacteria/host co-culture experiments, as previously described [10]. Cells were seeded at a density of  $2 \times 10^4$  cells per well in 48-well plates and cultured at 37°C/5% CO<sub>2</sub>, in growth media consisting of aMEM (Thermo-Fisher, Victoria, Australia) supplemented with 10% v/v foetal calf serum (FCS) (Thermo-Fisher), 2 mM L-Glutamine (Thermo-Fisher), 1 U/ml penicillin/streptomycin (Thermo-Fisher). At 90% confluence, SaOS2 cultures were switched to osteogenic differentiation medium comprising aMEM supplemented with 5% v/v FCS, 100 mM ascorbate 2-phosphate (Sigma-Aldrich, St Louis, USA), 1.8 mM potassium di-hydrogen phosphate (Sigma-Aldrich), 1 mM HEPES (Thermo-Fisher), 2 mM L-Glutamine and 1 U/ml penicillin/streptomycin. Cells were then maintained under differentiation conditions for 28 days to achieve osteocyte-like phenotype (SaOS2-OY) [9].

### *S. aureus* preparation

Two pre-characterised *S. aureus* strains, an MRSA strain, WCH-SK2 (SK2) [12] and an MSSA strain, WCH-SK3 (SK3), were grown in Terrific Broth (Thermo-Fisher) on a 37°C/200 rpm rocking platform to achieve log-phase suspension cultures individually. Bacteria were pelleted by 3000 g for 5 min centrifugation and then resuspended in sterile phosphate-buffered saline (PBS) to estimate cell number by the optical density (OD) at 600 nm light absorption. The two bacteria suspensions were then taken to undergo 10-fold serial dilutions and the diluents were individually plated on blood agar plates containing 10% v/v defibrinated sheep blood (Thermo-Fisher) to determine the colony-forming-unit (CFU) number from original bacterial suspension preparations. The calculated CFU numbers from agar plates were compared to the ddPCR quantified genome copy numbers.

### Host cell infection by co-culturing of SaOS2-OY and *S. aureus*

For host cell infection, bacterial suspension cultures were resuspended in sterile PBS, as described above, to achieve the density for low and high multiplicities of infection (MOI). The differentiation media for SaOS2-OY was removed and the cells washed twice with PBS. The resulting bacterial inoculums were added to SaOS2-OY cells to allow invasion for 1 hour at 37°C/5% CO<sub>2</sub>. Post-infection, cultures were washed twice with PBS and incubated at 37°C/5% CO<sub>2</sub> with 20 µg/ml lysostaphin (Sigma-Aldrich) in antibiotic-free media for 24 h to eliminate extracellular bacteria before replenishment with fresh differentiation medium. Supernatants were streaked on agar plates to verify the absence of extracellular bacteria at 24 h and 120 h post-infection.

### Measurements of intracellular bacterial number by CFU counting

The intracellular bacteria numbers were determined at 24 h and 120 h post-infection time points by spreading serial dilutions of cell lysates on blood agar plates and culturing at 37°C/5% CO<sub>2</sub> for 48 hours.

### DNA extraction from host cells and bacteria

For genomic quantification, total DNA was isolated using either DNeasy Blood & Tissue Kits (Qiagen Inc., VIC, Australia), or DirectPCR™ Lysis Reagent (Direct buffer) (Viagen Biotech Inc., CA, USA), as per the manufacturers' instructions. For the new approach of using the Direct buffer, 500 µl buffer together with 200 mg/ml proteinase K (Thermo-Fisher) was used for the lysis of one SaOS2-OY sample in one well from a 48-well tissue culture plate or one pelleted bacterial sample

prepared from serial dilutions of a suspension culture. Individual cell lysates were transferred to 1.5 ml reaction tubes, digested at 55°C for 35 min and then heat-inactivated at 85°C for 15 min to terminate enzymic digestion. The processed lysates were then ready for PCR analyses.

## Digital droplet polymerase chain reaction (ddPCR)

The ddPCR assay was performed using QX200 Digital Droplet PCR System (Bio-Rad Laboratories, USA), as per the manufacturer's instructions. Primer sets targeting a human genome-specific sequence within the single-copy type X collagen (*COL10A1*) gene and a *S. aureus* genome-specific sequence within the sigma factor B (*sigB*) gene, were used. The sequences of primer sets were:

*COL10A1*: forward 5'-ccaccaggtcaagcagtcac-3', reverse 5'-gttggcactaacaagaggggt-3'

*sigB*: forward 5'-ggggcaacaagatgaccatt-3', reverse 5'-tgccgttctctgaagtcgtg-3'

## Analysis of human bone tissues

All human studies received institutional research ethics approval (Royal Adelaide Hospital Human Research Ethics Committee Approval No. 14466). Bone biopsies were collected from patients undergoing either primary total hip replacement or revision total hip replacement surgery for PJI, with informed, written patient consent. Each bone specimen was separated into two parts. One of these was homogenised in a bead-beating tissue homogeniser (Bead Ruptor Elite, Omni International, Kennesaw, GA, USA), with 2 cycles of beating at 3 m/s for 30 s to disassociate the bone structure for the release of potentially viable bacteria. Bone homogenates were then sent to a clinical pathology laboratory (SA Pathology, Adelaide, Australia) for CFU analysis. Clinical soft tissue and fluid samples were sent to SA Pathology directly from surgery for pathogen analysis. The second piece of bone biopsy was processed and de-mineralised using OSTEOSOFT® (Sigma-Aldrich) solution for standard paraffin embedding procedure. Post-embedding, the bone tissues were sectioned under DNase/RNase-free conditions. One 5µm section from each bone specimen was used for Masson's Trichrome staining, as described in our previous study [18], and six sections were pooled together in DirectPCR™ Lysis Reagent for the isolation of total genomic DNA, as described above. Prepared DNA samples were then subjected to 1) conventional PCR amplification for an ONT sequencing readout (MinION sequencer, Oxford Nanopore Technology, Oxford, UK), according to the manufacturer's instructions; 2) ddPCR for the absolute quantification for both human and bacterial genome copy number. The primer sets targeting the human *COL10A1* genomic sequence, as listed above, and bacterial *tuf* sequence primers (forward 5'-tttcaatcactgtcgtgg-3', reverse 5'-ggagtatgacgtccacctc-3') were used for measuring human and bacterial genome copies, respectively. The output sequencing data were analysed by Epi2ME software within the WIMP workflow (Oxford Nanopore Technologies) for bacterial taxonomical analysis.

## Statistics

The results were graphed as means ± standard errors of the means (SEMs). The significance between two treatment groups was evaluated using two-tailed T-test using GraphPad Prism 10.2 (GraphPad Software, MA, USA), where applicable.

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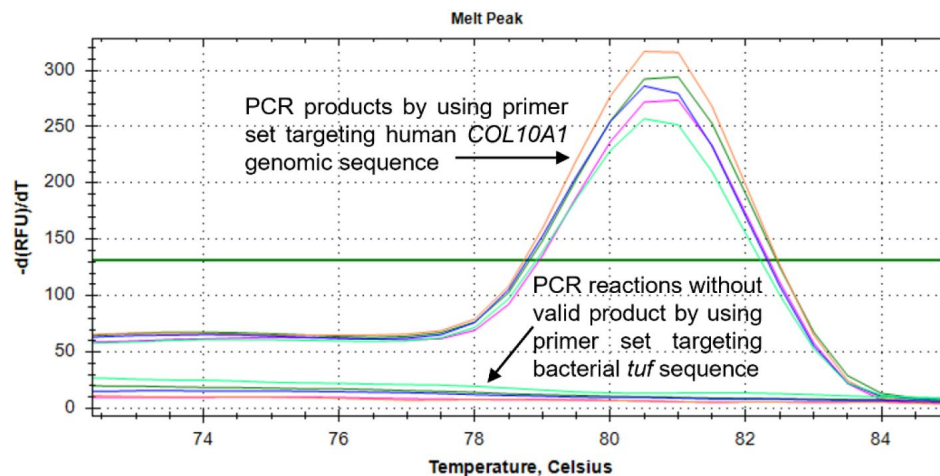
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## Author contributions

D.Y., G.J.A. and L.B.S. conceived the study. D.Y. adapted key protocols and developed the workflow; Q.S. performed most of the experimental work; K.H. contributed to molecular analysis and D.M. contributed to histological analysis; N.J.G and A.R.Z validated the methods developed in this work; Q.S. and D.Y. drafted the manuscript; G.J.A and D.Y provided scientific insight and supervised the study; L.B.S. consented patients and performed biopsy retrieval; all authors contributed to manuscript preparation and data presentation and agreed on the submitted version of the manuscript.

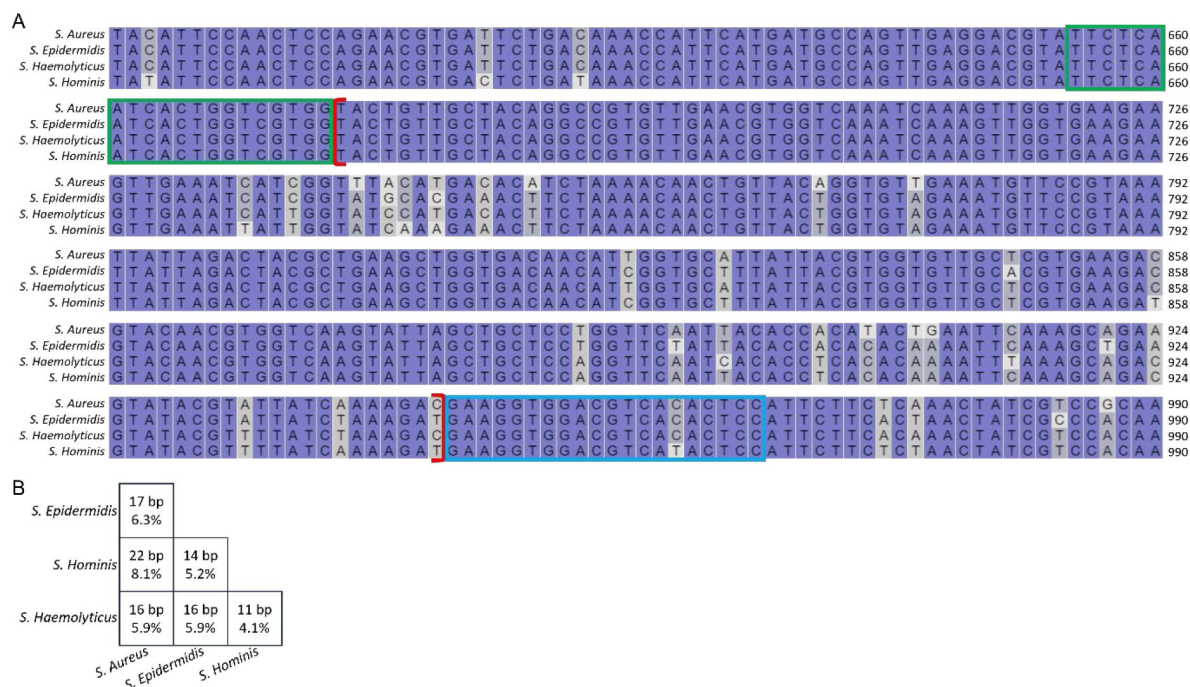
## Competing interest

The authors declare no competing interests.



## Supplementary figure 1

PCR analysis of bone samples from primary total hip replacement cases. DNA isolated from 5 patient bone samples were analysed by PCR for the presence of human *COL10A1* and bacterial *tuf*. The negative presence of bacterial *tuf* PCR product was confirmed by melt analysis post-quantitative PCR reactions using the DNA samples of five primary total hip replacement (non-infected) patients (coded with five different colours).



## Supplementary figure 2

Multi-sequence alignment among *S. aureus*, *S. epidermidis*, *S. haemolyticus* and *S. hominis*. The green and blue boxes indicate the forward and reverse primer binding areas, respectively; and the region between the red brackets was the assaying sequence for analysis (A). A summary of the number of mismatched base pairs and % differences within the assayed regions, among the above four species (B).

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**Reviewer #1 (Public Review):**

Summary:

This work shows, based on basic laboratory investigations of in vitro grown bacteria as well as human bone samples, that conventional bacterial culture can substantially underrepresent the quantity of bacteria in infected tissues. This has often been mentioned in the literature, however, relatively limited data has been provided to date. This manuscript compares culture to a digital droplet PCR approach, which consistently showed greater levels of bacteria across the experiments (and for two different strains).

Strengths:

Consistency of findings across in vitro experiments and clinical biopsies. There are real-world clinical implications for the findings of this study.

Weaknesses:

No major weaknesses. Only 3 human samples were analyzed, although the results are compelling.

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**Reviewer #2 (Public Review):**

In this study, the authors address discrepancies in determining the local bacterial burden in osteomyelitis between that determined by culture and enumeration by DNA-directed assay. Discrepancies between culture and other means of bacterial enumeration are long established and highlighted by Staley and Konopka's classic, "The great plate count anomaly" (1985). Here, the authors first present data demonstrating the emergence of discrepancies

between CFU counts and genome copy numbers detected by PCR in *S. aureus* strains infecting osteocyte-like cells. They go on to demonstrate PCR evidence that *S. aureus* can be detected in bone samples from sites meeting a widely accepted clinico-pathological definition of osteomyelitis. They conclude their approach offers advantages in quantifying intracellular bacterial load in their in vitro "co-culture" system.

#### WEAKNESSES

(A) My main concern here is the significance of these results outside the model osteocyte system used by this group. Although they carefully avoid over-interpreting their results, there is a strong undercurrent suggesting their approach could enhance aetiologic diagnosis in osteomyelitis and that enumeration of the infecting pathogen might have clinical value. In the first place molecular diagnostics such as 16S rDNA-directed PCR are well established in identifying pathogens that don't grow. Secondly, it is hard to see how enumeration could have value beyond in vitro and animal model studies since serial samples will rarely be available from clinical cases.

(B) I have further concerns regarding interpretation of the combined bacterial and host cell-directed PCRs against the CFU results. Significance is attached to the relatively sustained genome counts against CFU declines. On the one hand it must be clearly recognised that detection of bacterial genomes does not equate to viable bacterial cells with potential for further replication or production of pathogenic factors. Of equal importance is the potential contribution of extracellular DNA from lysed bacteria and host cells to these results. The authors must clarify what steps, if any, they have taken to eliminate such contributions for both bacteria and host cells. Even the treatment with lysotaphin may have coated their osteocyte cultures with bacterial DNA, contributing downstream to the ddPCR results presented.

#### STRENGTHS

(C) On the positive side, the authors provide clear evidence for the value of the direct buffer extraction system they used as well as confirming the utility of ddPCR for quantification. In addition, the successful application of MinION technology to sequence the EF-Tu amplicons from clinical samples is of interest.

(D) Moreover, the phenomenology of the infection studies indicating greater DNA than CFU persistence and differences between the strains and the different MOI inoculations are interesting and well-described, although I have concerns regarding interpretation.

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#### Author response:

The following is the authors' response to the original reviews.

##### **Reviewer #1 (Public Review):**

##### *Summary:*

*This work shows, based on basic laboratory investigations of invitro-grown bacteria as well as human bone samples, that conventional bacterial culture can substantially underrepresent the quantity of bacteria in infected tissues. This has often been mentioned in the literature, however, relatively limited data has been provided to date. This manuscript compares culture to a digital droplet PCR approach, which consistently showed greater levels of bacteria across the experiments (and for two different strains).*

### Strengths:

*Consistency of findings across in vitro experiments and clinical biopsies. There are real-world clinical implications for the findings of this study.*

### Weaknesses:

*No major weaknesses. Only three human samples were analyzed, although the results are compelling.*

We only put in three examples of clinical diagnosis to showcase the application of this method particularly to osteomyelitis. For further validation, larger cohort studies are required, which are currently underway.

### Reviewer #2 (Public Review):

*In this study, the authors address discrepancies in determining the local bacterial burden in osteomyelitis between that determined by culture and enumeration by DNA-directed assay. Discrepancies between culture and other means of bacterial enumeration are long established and highlighted by Staley and Konopka's classic, "The great plate count anomaly" (1985). Here, the authors first present data demonstrating the emergence of discrepancies between CFU counts and genome copy numbers detected by PCR in *S. aureus* strains infecting osteocyte-like cells. They go on to demonstrate PCR evidence that *S. aureus* can be detected in bone samples from sites meeting a widely accepted clinicopathological definition of osteomyelitis. They conclude their approach offers advantages in quantifying intracellular bacterial load in their in vitro "co-culture" system.*

The publication related to "The great plate count anomaly (1985)" has been added to revised version as new reference #2.

### Weaknesses

*- My main concern here is the significance of these results outside the model osteocyte system used by this group. Although they carefully avoid over-interpreting their results, there is a strong undercurrent suggesting their approach could enhance aetiologic diagnosis in osteomyelitis and that enumeration of the infecting pathogen might have clinical value. In the first place, molecular diagnostics such as 16S rDNA-directed PCR are well established in identifying pathogens that don't grow. Secondly, it is hard to see how enumeration could have value beyond in vitro and animal model studies since serial samples will rarely be available from clinical cases.*

Indeed, we initiated this study for the purpose of trying to improve the diagnostic outcomes for osteomyelitis, in particular that associated with prosthetic joint infection (PJI) but also all other forms, as the current gold-standard diagnostic approaches for this type of infection, either bacterial culture or whole genome sequencing, are very time consuming and costly, and yet are not necessarily accurate. Our method has the benefits (not limited to) of achieving absolute quantification of bacterial load in a shortened time period (in the order of hours) in clinical bone specimens from infected patients. Many of the identified bacterial species in patients were not able to be diagnosed by standard bacterial culturing. Moreover, one of the problematic features of treating bone infection is that repetitive surgeries are usually needed, particularly in PJI, hence, serial clinical bone specimens from the same patient are in fact often available. Therefore, our method of being able to quantify bacterial load offers the advantage of monitoring the infected status throughout the treatment journey. In this study, we chose the *tuf* gene as the targeting sequence to amplify the bacterial signal instead of the well-established 16S PCR for the reason that *tuf* provides much better sequence

discrimination between bacterial species. Therefore, the short PCR amplicon of just 271 bp used in our study, is able to give us a highly accurate taxonomic readout. By this approach, we again shorten the time required for diagnosis. In the last paragraph of the Discussion in the revised manuscript, extra text, a figure demonstrating the strong sequence diversity in *tuf* (Supplementary Figure 2) and an additional reference have been added to address the Reviewer's concerns.

*- I have further concerns regarding the interpretation of the combined bacterial and host cell-directed PCRs against the CFU results. Significance is attached to the relatively sustained genome counts against CFU declines. On the one hand, it must be clearly recognised that the detection of bacterial genomes does not equate to viable bacterial cells with the potential for further replication or production of pathogenic factors. Of equal importance is the potential contribution of extracellular DNA from lysed bacteria and host cells to these results. The authors must clarify what steps, if any, they have taken to eliminate such contributions for both bacteria and host cells. Even the treatment with lysotaphin may have coated their osteocyte cultures with bacterial DNA, contributing downstream to the ddPCR results presented.*

We agree that concerns around the interpretation of any molecular readout need to be taken into account. We have yet to find a method that can definitively identify bacterial viability in a clinical setting in the absence of culture. However, PJI and osteomyelitis in general is characterised by a high percentage of culture-negative infection cases, calling for such molecular approaches. Commercially available, so called “live/dead” bacterial PCR reagents exist that act as PCR signal inhibitors by penetrating the cell wall of compromised cells to prevent the PCR signal being generated from those cells. In our experience, while these can provide a certain level of added scrutiny in an experimental setting, they are not definitive because the reaction is often incomplete in an idealised situation and also the reagent may cancel signal from viable bacteria growing under conditions of stress, such as during antimicrobial treatment and host-derived stress imparted in intracellular or intra-tissue environments. Indeed, such stresses are likely contributors to clinical non-culturability. Whole genome sequencing would provide more certainty of bacterial viability to demonstrate genomic intactness but as we discuss herein, this a lengthy and costly process, and one which may prove difficult from host tissue with a low pathogen load. It should be noted that the significance of any diagnostic readout, including from culture, WGS or our method reported here would need to be interpreted by the treating clinical team. We would argue that a rapid, practical molecular diagnostic method in the absence or even presence of culture would provide treating clinicians with an improved rationale for tailoring antimicrobial treatments.

#### *Strengths*

*- On the positive side, the authors provide clear evidence for the value of the direct buffer extraction system they used as well as confirming the utility of ddPCR for quantification. In addition, the successful application of MinION technology to sequence the EF-Tu amplicons from clinical samples is of interest.*

*- Moreover, the phenomenology of the infection studies indicating greater DNA than CFU persistence and differences between the strains and the different MOI inoculations are interesting and well-described, although I have concerns regarding interpretation.*

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